

———— // ZPWR-CLIP-ENGINE // ————

zpwr-clip-engine

v0.1.0

REFERENCE MANUAL

one arranger, every app, one grid – content swapped, never re-implemented

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MenkeTechnologies

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Overview

zpwr-clip-engine is the shared DAW arranger / sequencer for the MenkeTechnologies GUI app stack. It is one canvas-2D grid with swappable domains - arranger, notes/piano-roll, launcher/session, automation lanes - plus the clip palette, MIDI export, and project/bank serialization. The real scheduling engine lives here too ([engine/](#)): the pattern -> events -> step/transport/MIDI scheduler, extracted from zpwr-daw as a pure-C++17, header-only C ABI ([zpc_clip_*](#)) - header-only [ClipEngine](#) + [MidiFile](#), no JUCE, no external deps. Created by MenkeTechnologies.

- One grid, swappable domains. A single canvas grid renders arranger, piano-roll, launcher/session, and automation-lane domains - content is swapped, never re-implemented per app.
- One engine, every consumer. The [zpc_clip_*](#) C ABI builds two ways from one source: [libzpwr_clip_engine_static.a](#) (the daw links it natively) and [libzpwr_clip_engine.dylib/.so](#) (the Tauri apps load it through a Rust FFI wrapper).
- Mount through one entry. Hosts call [initClipSeq\({ nf, getCAT }\)](#) and supply a backend; the [nf](#) functions bridge the grid to whichever engine the host uses.
- No JUCE, no deps. The engine is header-only C++17 ([ClipEngine](#) + [MidiFile](#)) with no JUCE and no external dependencies - so it embeds in native and Tauri hosts alike.
- Audio & video clips. Beyond MIDI patterns, tracks can hold audio clips (waveform + fades) and video clips (filmstrip + in/out trim). Video clips drop in-browser (frames sampled to a filmstrip), and a program monitor - dockable or popped out into a draggable, resizable player - previews the frame under the playhead. The foundation for a video-editing timeline.

Hosts

- `zpwir-daw` (JUCE / C++) - drives the grid through native `nf` functions backed by this engine plus its own audio voices / per-track instruments (real sound).
- Tauri apps (Audio-Maxor, traderview, ztranslator) - bridge `nf` to the C ABI via a Rust FFI wrapper.
- Domains - arranger, notes / piano-roll, launcher / session, automation lanes.
- Clip palette - the swappable clip bank shared across domains.
- MIDI export - pattern/event data written out via the engine's `MidiFile`.
- Serialization - project / bank save & load.

Mounting

A host mounts the shared grid through one entry point and supplies its backend bridge:

```
import { initClipSeq } from './webui/clip/clip-seq.js';

// nf = the host's bridge functions (native in the daw, FFI in Tauri apps);
// getCAT = the host's content/category provider for the clip palette
initClipSeq({ nf, getCAT });
```

The host page embeds the `clip.html` markup (a `#pane-clip` section with a `#clip-grid` canvas and the toolbar ids it documents), links `clip.css`, then:

```
import { initClipSeq } from './webui/clip/clip-seq.js';
// non-audio host (synth / fx / midi-fx / future GUI app):
const api = initClipSeq(window.makeBasicClipBackend());
// the daw passes its own native { nf, getCAT } instead.
```

`initClipSeq(deps)` reads `deps.nf` and `deps.getCAT` (and optionally `deps.el`, `deps.onSelectTrack`).

Layout

```
webui/  
  clip/  
    clip-seq.js      initClipSeq(deps) -- the mount: arranger/session/clips + drill-in  
    clip.js          initClip(deps) -- single-grid clip editor (the simpler entry)  
    clip-grid.js     standalone single-pattern grid wiring  
    clip-bank.js     bank import/export modal (localStorage-backed)  
    clip-basic-backend.js window.makeBasicClipBackend() -- shared NON-AUDIO backend  
    clip.css         clip-pane styling  
    clip.html        #pane-clip markup partial (the host embeds this)  
    *-demo.html      standalone browser demos  
  grid/  
    grid-core.js     canvas renderer + layout/hit-test  
    grid-model.js    createModel -- cell store + localStorage persistence  
    grid-interactions.js pointer/drag/keyboard interactions  
    index.js         createGrid -- the bundled "all-in-one" factory  
    cursors.js       tool cursors  
    project.js       serialize/deserialize project + bank  
    als-import.js    Ableton Live Set (.als) -> project (float-time notes)  
    browser-drawer.js docked content browser (audio/MIDI/presets/plugins)  
    sequencer.js     slot sequencer + swing helpers  
    export/midi.js   patternToMidi / downloadMidi (Standard MIDI File)  
    domains/        notes, arranger, launcher, autolanes, automation, triggers  
    transport/       juce-bridge.js, tauri-bridge.js  
    tests/           *.test.mjs (node --test)  
engine/              the REAL backend (C ABI), extracted from zpw-r-daw -- no JUCE:  
  include/zpc/  
    ClipEngine.h     header-only pattern->events->step/transport scheduler  
    MidiFile.h       Standard MIDI File writer (backs render_midi)  
    capi/clip_engine.h C ABI: zpc_clip_new/free/set_pattern/set_transport/play/  
                      step/advance/poll_events/pending_events/render_midi  
  src/capi/clip_engine.cpp C ABI implementation over ClipEngine  
  CMakeLists.txt      libzpw_clip_engine.{dylib,so} (shared) + _static.a  
  
clip-seq.js imports its grid pieces with ../grid/... relative paths (individually, not via ../grid/index.js, to dodge  
a JUCE resource basename clash); the layout above preserves those paths so the imports resolve unchanged.
```

getCAT() contract

`clip-seq.js` reads exactly four fields off the catalog; the grid domains never touch it:

field	basic backend	meaning
<code>hasClipSeq</code>	<code>true</code>	a step sequencer exists (drives the <code>clip-view</code> clock)
<code>hasClipAudio</code>	<code>false</code>	native JUCE audio-clip engine wired (daw only)
<code>hasKeyboard</code>	<code>true</code>	playback is allowed (gate on <code>clipStart</code>)
<code>name</code>	<code>'clip'</code>	host id; the daw uses <code>'zpw1-daw'</code> (per-track instruments)

nf(name) contract

`nf(name)` returns a per-command function. `clip-seq.js` wraps almost every call in `try/catch` and persists pattern/arrangement/project state itself via `localStorage` (`zfx_clip_*`, `zfx_clip_arrangement`, ...), so most commands can be neutral resolves. The basic backend never throws and produces no audio.

command	daw (native)	basic backend
<code>clipSeqPattern</code>	loads the pattern into the audio sequencer	no-op resolve (UI persists to <code>localStorage</code>)
<code>clipSeqTransport</code>	sets steps/bpm/div/swing/loop on the engine	real - configures the JS step timer
<code>clipSeqPlay</code>	starts/stops the audio-thread clock	real - starts/stops a <code>setInterval</code> step timer
<code>clipSeqStep</code>	returns the audio clock's current step	real - returns the JS timer's current step
<code>clipSeqExportMidi</code>	writes the SMF via native file IO	real - decodes base64 SMF, browser download
<code>clipSeqSaveFile</code>	native file save (only when <code>window.Juce</code>)	no-op resolve
<code>clipAudioLoad</code>	loads an audio sample, returns peaks/dur	no-op resolve <code>null</code> (never reached)
<code>clipAudioTrigger</code>	fires a native audio voice	no-op resolve (never reached)
<code>clipAudioStopAll</code>	stops all audio voices	no-op resolve (never reached)
<code>setXfade</code>	sets the audio-clip crossfade	no-op resolve
<code>loadTrackInstrument</code>	loads a hosted VST3/AU into a track	no-op resolve
<code>dawSetTrackInstrument</code>	turns a track into a built-in voice synth (empty patch = default note-pitched voice)	no-op resolve
<code>dawClearTrackInstrument</code>	removes a track's built-in voice	no-op resolve
<code>trackPresetSave</code>	writes the active track's graphs to a named <code>.ztrk</code> (<code>{type,note?,stereo?}</code>)	no-op resolve
<code>trackPresetList</code>	lists the saved <code>.ztrk</code> presets (<code>[{name,type}]</code>)	no-op resolve <code>"[]"</code>
<code>trackPresetLoad</code>	reads a named <code>.ztrk</code> preset back	no-op resolve <code>""</code>
<code>trackPresetDelete</code>	deletes a named <code>.ztrk</code> preset	no-op resolve
<code>getPatch/setPatchJson</code>	reads/writes the track's note graph (preset store)	no-op resolve
<code>trkgetPatch/trksetPatchJson</code>	reads/writes the track's stereo graph (preset store)	no-op resolve
<code>midiNoteOn</code>	native MIDI note-on (audible)	no-op resolve (no synth attached)
<code>midiNoteOff</code>	native MIDI note-off	no-op resolve
<code>setActiveLayer</code>	selects the active synth layer	no-op resolve
<code>setTrackCount</code>	sizes the per-track audio graphs	no-op resolve
<code>clipAutoValue</code>	emits a live automation CC value	no-op resolve

Any command the UI adds later but the table omits falls through to a generic neutral resolver (resolve `undefined`), so

the host degrades rather than throws.

`makeBasicClipBackend()` is exposed both as `window.makeBasicClipBackend` (for the WebView hosts) and as a CommonJS export (for `node --test`).

Ableton Live Set (.als) import

`grid/als-import.js` reads an Ableton Live Set and returns a `serializeProject()`-shaped object, so the host feeds it straight through `deserializeProject` + `applyProjectState` - the same path as a native `.zdp`. This is the cross-DAW interop Bitwig ships: an Ableton arrangement comes in as tracks, clips, MIDI notes, tempo, time signature, arrangement placement, locators, and audio-clip sample paths. Devices/plugins, automation envelopes and audio warping have no target in the clip model and are dropped (as in any cross-DAW import).

`.als` is gzip XML; the host's `gunzipText` decompresses, then routes text starting with `<Ableton` to `alsToProject`. The parser is a dependency-free XML walker (no `DOMParser`, which is absent under Node) so it runs headless in CI. Verified against real sets spanning Live 8.2.1 through Live 12.2.

Float time. A note's exact start is encoded in its fractional grid-step key ("`12.5`" = step 12 1/2); its exact length rides in the optional `bl` field (beats). Integer keys + absent `bl` are the legacy step-sequencer shape, unchanged - old `.zdp` projects load byte-identically. The canvas renderer draws notes at the sub-cell offset and the JS clock micro-delays their note-on. The native step engine (`engine/include/zpc/ClipEngine.h`, the single-clip preview path) has no sub-step slot, so it rounds a fractional start to the nearest step; arrangement playback runs on the float-accurate JS clock.

Tests

The grid/domain unit tests live in `webui/grid/tests/` and run with the Node test runner:

```
node --test webui/grid/tests/
```

Architecture

zpw-clip-engine is consumed two ways from one source. The frontend (`webui/`) is mounted by hosts through `initClipSeq({ nf, getCAT })`: one canvas grid renders swappable domains (arranger, notes/piano-roll, launcher/session, automation lanes) plus the clip palette. The engine (`engine/`) is the scheduler as a pure-C++17, header-only C ABI (`zpc_clip_*`) - `ClipEngine` + `MidiFile`, no JUCE, no external deps - built both as a static lib the JUCE daw links natively and as a `dylib/.so` the Tauri apps load through a Rust FFI wrapper. The `nf` bridge is what differs per host; the grid and the scheduler do not.

Module map

area	responsibility
<code>webui/grid</code>	the canvas grid - model, interactions, cursors, sequencer, project state, domain entry (<code>index.js</code> , <code>grid-core.js</code> , <code>grid-model.js</code> , <code>grid-interactions.js</code> , <code>sequencer.js</code> , <code>project.js</code>)
<code>webui/clip</code>	the clip layer - clip grid, clip bank/palette, sequencer view, and the non-audio basic backend for preview (<code>clip.js</code> , <code>clip-grid.js</code> , <code>clip-bank.js</code> , <code>clip-seq.js</code> , <code>clip-basic-backend.js</code>)
<code>webui/lib</code>	shared frontend helpers used across the clip and grid layers
<code>engine/</code>	the C++17 header-only scheduler exposed as the <code>zpc_clip_*</code> C ABI - <code>ClipEngine</code> (pattern -> events -> step/transport) and <code>MidiFile</code> (MIDI export)
<code>bindings/</code>	the build glue producing the static lib (native daw link) and the <code>dylib/.so</code> (Tauri Rust-FFI load) from the one engine source

Consumers

1. zpw-daw (JUCE / C++) - links `libzpw_clip_engine_static.a` natively and drives the grid via native `nf` functions, adding its own audio voices / per-track instruments for real sound.
2. Tauri apps (Audio-Maxor, traderview, ztranslator) - load `libzpw_clip_engine.dylib/.so` through a Rust FFI wrapper and bridge `nf` to the C ABI.
3. One source of truth - the grid and scheduler are never re-implemented per app; only the `nf` backend bridge and supplied content differ.

Build

The engine builds two artifacts from one C++17 header-only source via `bindings/`: `libzpw_clip_engine_static.a` for native linking and `libzpw_clip_engine.dylib/.so` for FFI loading. Because it is header-only with no JUCE or external dependencies, the same engine embeds in native and Tauri hosts without divergence.